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NBA Player Replacement Tool

Using unsupervised learning to find statistical player twins

The Problem

A player gets injured. A trade happens. The front office needs to move fast.

The question isn't just 'find a good player.' It's 'find a player who fills the same statistical role.' A team that loses a starting point guard needs someone who distributes and pushes pace, not just anyone available.

This is hard because there are no ground truth labels. Player roles are subjective. And some players genuinely have no statistical twin.

[GitHub Link](#)

A Three-Model Pipeline

Each model answers a different question at a different stage of the same decision.

01



K-Means

*What type are
we replacing?*

Groups 539 players into 5 archetypes with no predefined labels.

02



NMF

*Who should
we call?*

Matrix factorization builds a ranked list of statistical twins.

03



DBSCAN

*Is replacement
even possible?*

Flags players with no statistical twin, before you waste time searching.

The Data

12,844

player-seasons
1996–2023

539

players modeled
most recent season

13

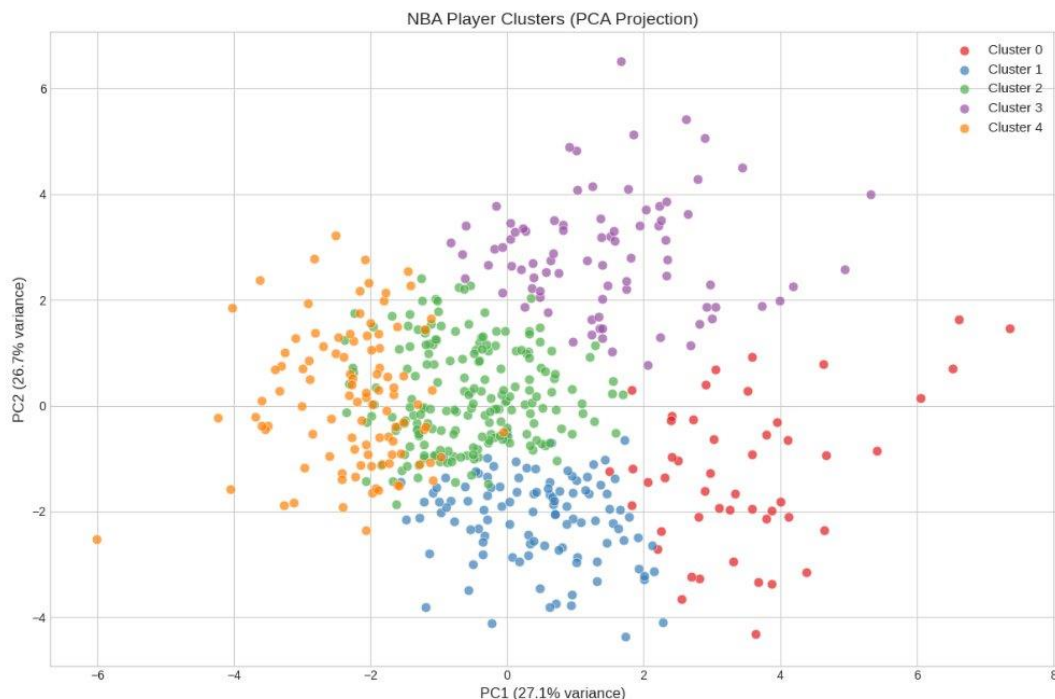
features
stats + bio

Features used

pts · reb · ast · net_rating · oreb_pct · dreb_pct · usg_pct · ts_pct · ast_pct · age · height · weight · gp

StandardScaler → K-Means | MinMaxScaler → NMF (non-negativity requirement)

K-Means: Results



cluster	0	1	2	3	4
age	26.24	26.50	26.62	27.27	23.91
player_height	209.63	207.49	196.93	193.83	194.43
player_weight	113.12	107.59	95.27	92.27	89.51
gp	64.65	40.78	57.02	61.01	19.97
pts	15.38	5.35	8.33	19.31	3.26
reb	8.76	3.60	2.86	4.60	1.42
ast	2.74	0.83	1.65	5.44	1.10
net_rating	1.33	-2.75	0.54	0.79	-7.60
oreb_pct	0.09	0.08	0.03	0.02	0.03
dreb_pct	0.22	0.18	0.10	0.12	0.11
usg_pct	0.20	0.15	0.16	0.25	0.16
ts_pct	0.62	0.60	0.58	0.57	0.44
ast_pct	0.14	0.08	0.11	0.26	0.15

What the numbers show:

- Cluster 3 (Star Playmaker) — highest pts, ast, usg
- Cluster 0 (PF/C) — tallest, heaviest, most rebounds
- Cluster 4 (Young/Dev) — fewest games, lowest efficiency

K-Means: 5 Player Archetypes

	Power Forward / Center	15.4 pts	8.8 reb	2.7 ast	n=49
	Backup Big / Fringe Roster	5.4 pts	3.6 reb	0.8 ast	n=109
	Role Player / Bench Wing	8.3 pts	2.9 reb	1.7 ast	n=195
	Star Playmaker	19.3 pts	4.6 reb	5.4 ast	n=84
	Young / Developmental	3.3 pts	1.4 reb	1.1 ast	n=102

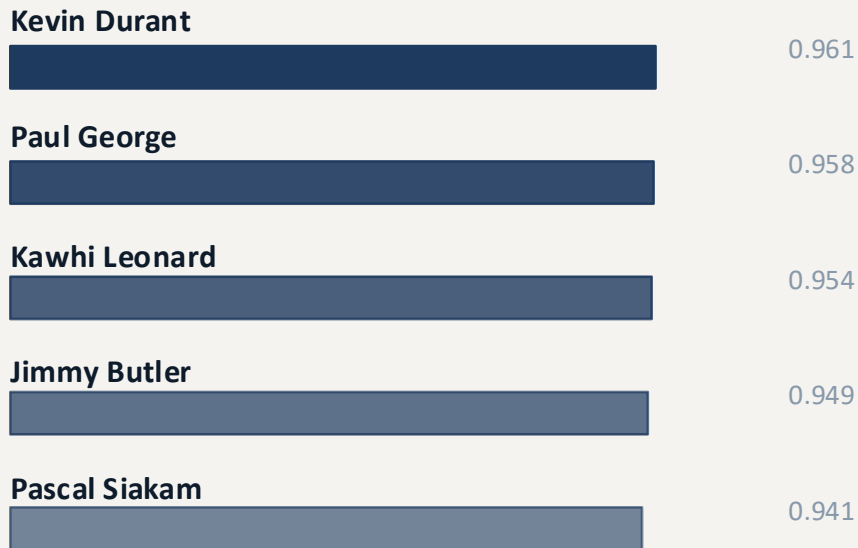
k=5 chosen over silhouette peak at k=3 (0.202) — k=3 produced overly broad groupings with little basketball interpretability.

NMF Recommender

Matrix factorization → latent skill components → cosine similarity → ranked list of twins

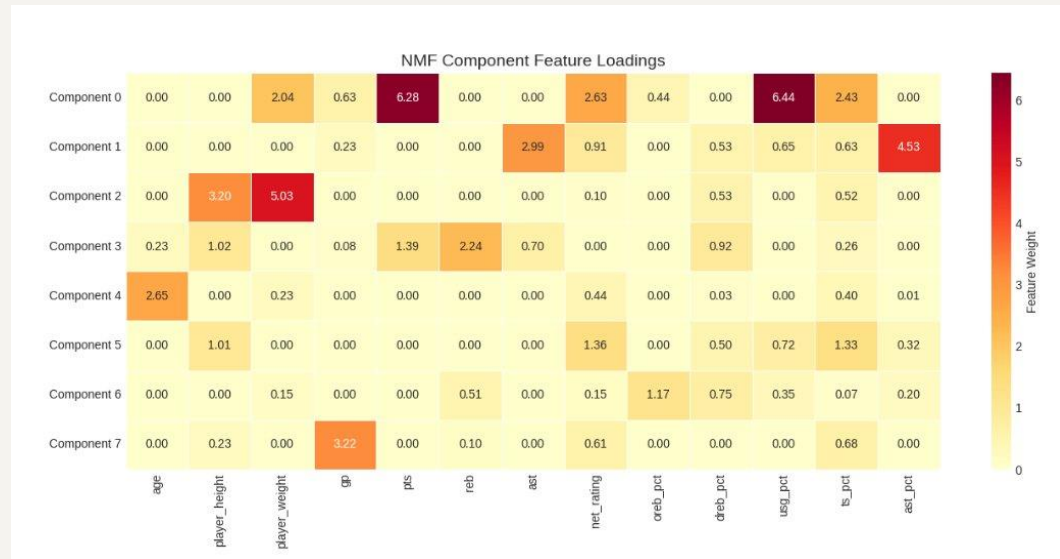
- 1 MinMaxScale the player-stats matrix
- 2 NMF decomposes into 8 latent skill components
- 3 Each player gets a representation vector
- 4 Cosine similarity ranks every player vs every other
- 5 Output: ranked list of statistical twins

Most similar to LeBron James



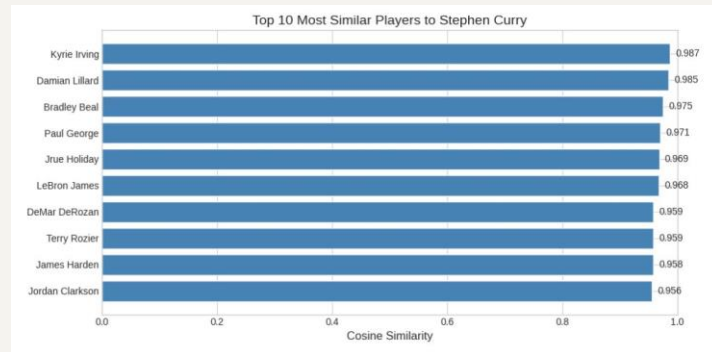
NMF: Results

Latent Component Loadings

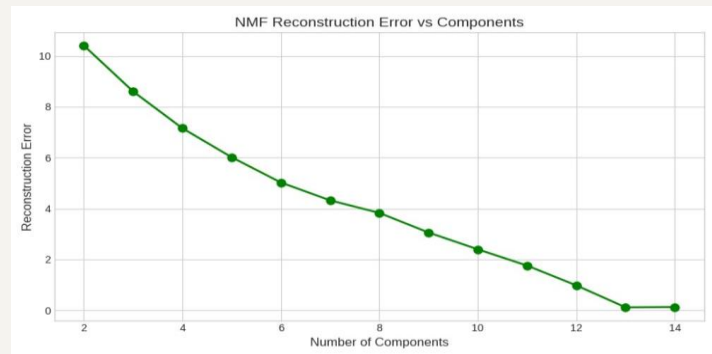


Component 0 captures scoring load (pts, usg_pct). Component 2 captures frontcourt presence (height, weight). Component 1 captures playmaking (ast_pct). NMF learned these dimensions without being told.

Example: Stephen Curry replacements



Component selection (elbow)



DBSCAN: Who Can't Be Replaced?

51

outliers
9.5% of players

Elite Outliers

Embiid · Luka · Giannis · LeBron · Jokic

No one else combines their volume + efficiency. No true replacement exists.

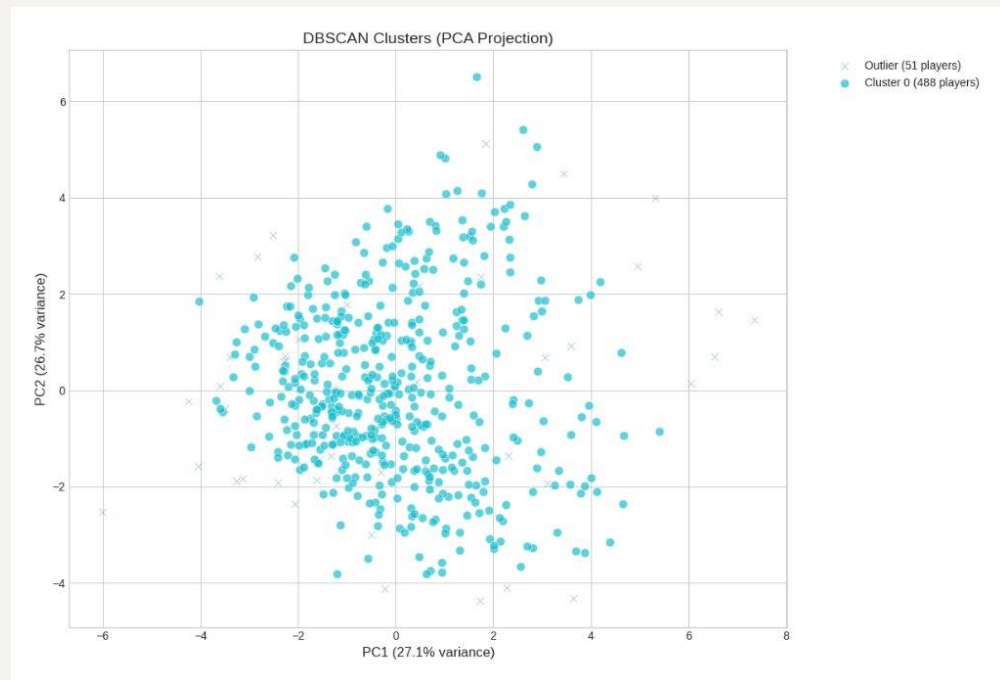
Fringe Outliers

Near-zero stats across the board. Too few minutes to cluster with anyone.
Statistically isolated, not uniquely skilled.

If a front office loses an elite outlier, the NMF recommender still returns names, but the similarity scores tell the real story.

DBSCAN: Results

Cluster vs Outlier (PCA projection)



Top outliers flagged

=== Players Flagged as Outliers by DBSCAN ===

player_name	pts	reb	ast	usg_pct	ts_pct	archetype
Joel Embiid	33.1	10.2	4.2	0.370	0.655	Power Forward / Center
Luka Doncic	32.4	8.6	8.0	0.368	0.609	Star Playmaker / Primary Ball Handler
Giannis Antetokounmpo	31.1	11.8	5.7	0.373	0.605	Power Forward / Center
LeBron James	28.9	8.3	6.8	0.322	0.583	Star Playmaker / Primary Ball Handler
Zion Williamson	26.0	7.0	4.6	0.288	0.652	Power Forward / Center
Nikola Jokic	24.5	11.8	9.8	0.263	0.701	Power Forward / Center
James Harden	21.0	6.1	10.7	0.247	0.607	Star Playmaker / Primary Ball Handler
Tyrese Haliburton	20.7	3.7	10.4	0.238	0.624	Star Playmaker / Primary Ball Handler
Louis King	20.0	4.0	2.0	0.205	0.744	Role Player / Bench Wing
Cade Cunningham	19.9	6.2	6.0	0.287	0.492	Star Playmaker / Primary Ball Handler
Domantas Sabonis	19.1	12.3	7.3	0.213	0.668	Power Forward / Center
RaiQuan Gray	16.0	9.0	7.0	0.205	0.621	Power Forward / Center
Brook Lopez	15.9	6.7	1.3	0.187	0.630	Power Forward / Center
Mac McClung	12.5	5.0	4.5	0.243	0.563	Star Playmaker / Primary Ball Handler
Kevin Love	8.2	6.4	1.9	0.185	0.548	Power Forward / Center
Jay Huff	7.3	3.0	1.4	0.155	0.796	Backup Big / Fringe Roster
Chance Comanche	7.0	3.0	0.0	0.140	0.518	Young / Developmental Player
Jay Scrubb	6.5	3.0	0.5	0.151	0.825	Backup Big / Fringe Roster
Andre Drummond	6.0	6.6	0.5	0.196	0.606	Power Forward / Center
Kenneth Lofton Jr.	5.0	2.1	0.8	0.267	0.572	Backup Big / Fringe Roster
Michael Carter-Williams	4.3	1.3	1.8	0.191	0.498	Young / Developmental Player
Moses Brown	4.3	3.9	0.1	0.188	0.607	Backup Big / Fringe Roster
Xavier Sneed	4.3	1.3	1.3	0.136	0.723	Young / Developmental Player
Udonis Haslem	3.9	1.6	0.0	0.185	0.433	Backup Big / Fringe Roster
Boban Marjanovic	3.3	1.9	0.3	0.206	0.710	Backup Big / Fringe Roster
Jacob Gilyard	3.0	4.0	7.0	0.052	0.500	Young / Developmental Player
Tyler Dorsey	3.0	0.7	0.0	0.227	0.900	Role Player / Bench Wing
Lester Quinones	2.5	0.8	0.5	0.283	0.654	Young / Developmental Player
Neemias Queta	2.4	2.2	0.2	0.141	0.607	Backup Big / Fringe Roster
Chris Silva	2.0	0.0	0.0	0.250	1.000	Role Player / Bench Wing
Stanley Umude	2.0	0.0	0.0	0.500	0.532	Role Player / Bench Wing
Donovan Williams	2.0	1.0	0.0	0.385	0.400	Young / Developmental Player
Ryan Rollins	1.9	1.0	0.5	0.260	0.508	Young / Developmental Player
Xavier Moon	1.8	0.8	1.3	0.233	0.389	Young / Developmental Player
Dylan Windler	1.7	0.0	0.3	0.182	0.833	Role Player / Bench Wing
Jordan Schakel	1.5	0.0	0.5	0.154	0.750	Role Player / Bench Wing
Facundo Campazzo	1.3	0.3	1.1	0.144	0.360	Young / Developmental Player
Braxton Key	1.3	0.3	0.0	0.091	1.064	Role Player / Bench Wing
Chima Moneke	1.0	1.0	0.5	0.250	0.347	Young / Developmental Player
Malcolm Hill	1.0	0.6	0.0	0.167	0.625	Role Player / Bench Wing

X markers = outliers. Dense core = 488 players who cluster normally. Stars scatter at the edges — their profiles have no near neighbors.

How the Models Compare

	K-Means	NMF	DBSCAN
Output	Discrete archetype	Similarity score	Cluster or outlier
Primary question	What type am I replacing?	Who should I call?	Is replacement possible?
Covers all players?	Yes	Yes	Flags 9.5% as outliers
Best for	Archetype classification	Finding twins	Irreplaceability detection

These models aren't competing, they're sequential. They answer different questions at different stages of the same decision.

Conclusions

The pipeline works.

K-Means identifies archetypes. NMF ranks statistical twins. DBSCAN flags the irreplaceable.

NBA talent is a spectrum, not buckets.

Silhouette peak of 0.202 and a single DBSCAN cluster both confirm it: the models are useful approximations of a continuous space.

This is a first filter, not a final answer.

Salary, age curve, contract, and team fit aren't here. Start with the data. Then watch the tape.